From Experimental Histories to Causal Surrogates: Data Reuse, Optimization, and Error Propagation*

Anonymous authors
Paper under double-blind review

Working manuscript, 8 October 2026. Archived attribution and physical validation are complete. The frozen independent-system experiment and its original independent audit are complete. Supplemental reproduction qualification, prospective result integration and final scientific review remain pending.

Abstract

A sequential experimental learner need not be the best model delivered after collection ends. We investigate whether retaining and refitting a completed experimental history improves the resulting causal surrogate, and which parts of the delivery procedure account for that change. A completed confirmation on one fixed deterministic emulator covers 12 acquisition histories: the median snapped exact-level error over three scored refit initializations on all eligible paid observations is 0.183 times the online learner's error in geometric mean across histories (95% interval 0.102–0.327), with 11 histories improving. This comparison combines data reuse and additional optimization; it does not identify an architecture effect or an advantage at equal fitting compute. An exploratory attribution study on the same exposed grid gives continuous-error ratios of 0.551 versus a development-selected flat refit and 0.423 versus a matched-CPU flat fit, using fixed initialization zero. We give conditional statements for mechanism eligibility, deterministic error propagation and quantization sensitivity, with explicit counterexamples to monotonic improvement. An archived physical case study retains all eleven conditions: delivery beats rolling learning in 7 conditions but Fourier regression in 0. The frozen independent-system comparison has completed its original audit; its results await supplemental reproduction qualification and are not reported in this version. The present evidence supports a scoped post-collection delivery finding, without an acquisition or foundation-model claim.

1 Introduction

An experimental learner has two jobs: maintain a useful model while observations arrive, and deliver a predictor after collection ends. These jobs can favor different uses of the same paid observations. A rolling buffer limits update cost, but it also forgets earlier parts of the experimental support. A completed history permits a fresh batch fit, whose extra computation is spent after the experimental responses have already been obtained. Evaluating only the online learner can therefore understate the predictive value of the collected data. Conversely, an improvement after refitting can be wrongly credited to causal factorization when ordinary data retention or optimization explains it.

We study this distinction for structural causal model (SCM) surrogates with a known graph and measured intermediate variables. A mechanism network is fit on observed parents, but a deployed chain must use its own predicted intermediates. Thus a local regression improvement and a downstream prediction improvement are different outcomes. Intervention masks determine which observations can estimate each mechanism, and the joint parent support determines where such estimates are constrained. These details matter even when every measured variable lies inside its training range.

*AI-assisted tools supported research planning, code development, analysis and manuscript drafting. The human authors are responsible for the final scientific assessment, verification and submission.

Our central question is empirical: *what does a completed experimental history buy at delivery time, and what accounts for that benefit?* We investigate three explanations: retaining observations that a rolling buffer discards, using paid observations that never entered a live update, and performing more optimization. A direct flat predictor and fixed classical regressors test whether a simpler refit can explain the result. We use "delivery" to denote this post-collection fitting procedure, without implying a new optimization algorithm or a general identification guarantee.

Contributions. First, we document a completed comparison over repeated histories of one fixed emulator and an exploratory factorial attribution with unchanged original weights and explicit fitting costs. Second, we formalize the eligibility of interventional observations for mechanism regression and explain the distinction between observed-parent fit and free-running prediction through propagation bounds and support counterexamples. These elementary results clarify the empirical estimand and its limits; they are not new general causal-identification theorems. Third, we report an archived physical prediction study with domain controls and all conditions retained, including unfavorable results. A frozen prospective study tests the selected recipe on independent parameterizations and reserved action blocks. Its original audit is complete; prospective results remain unreported here pending supplemental reproduction qualification and integrated evidence review.

Within the scratch-fit matrix, long fits on retained histories outperform final-buffer fits and 100-update fits; adding unused paid rows changes aggregate continuous error much less. These conditional recipe contrasts do not partition the original online-to-delivery gain. The matrix also favors the structured recipe against the selected flat controls on the exposed evaluation grid. Differences in usable rows, intermediate supervision and computation limit that interpretation. The physical study supplies a complementary boundary: a domain-informed Fourier predictor outperforms neural delivery in every evaluated condition. Our contribution is therefore an assessment of delivery behavior and accounting, with its scope fixed by the controls and evaluation design.

2 Setting and delivery procedure

Let G be a known DAG, $\mathrm{Pa}(i)$ the parents of node i, and R its root nodes. A paid history is a sequence $\mathcal{H} = \{(a_t, v_t)\}_{t=1}^{T}$, where a_t specifies the intervention set I_t and its clamped values, and v_t records all measured nodes. A response is counted when obtained, including duplicate commands and probes that were not chosen for an online update. Observation reuse adds fitting computation, not experimental responses.

For an invariant deterministic mechanism $v_i = f_i(v_{\mathrm{Pa}(i)})$, define its eligible rows by $E_i(\mathcal{H}) = \{t : i \notin I_t\}$. With a retained row set S, the delivery head minimizes the empirical objective

$$\widehat{\mathcal{L}}_{i,S}(\theta_i) = \frac{1}{|S \cap E_i|} \sum_{t \in S \cap E_i} \big(\hat{f}_{i,\theta_i}(v_{t,\mathrm{Pa}(i)}) - v_{t,i}\big)^2. \tag{1}$$

The objective requires $|S \cap E_i| > 0$; an empty eligible set supplies no fitted mechanism and blocks a delivery claim for that head. This objective uses measured parents during fitting. At deployment, each non-clamped node is evaluated in topological order at predicted parents; clamped nodes take their specified values. We call this *free-running* prediction. For noisy systems, conditional-mean regression is justified only under the assumptions in Section 8; recursively composing such means is not generally an interventional-mean estimator.

Recipe and simpler control. The selected delivery recipe fits separate two-hidden-layer ReLU networks of width 64 from scratch, with full-batch Adam, learning rate 0.002, and 30,000 updates per head. Initialization zero is the deployment specification. A 100-update version is the optimization ablation. The simpler control uses the same MLP form for a direct root-to-target regression. It predicts the composed map without supplying intermediate labels. In the archived attribution, intermediate-intervention rows are excluded from this flat regression because its root-only inputs cannot represent those responses. In the prospective root-only design, all training rows are eligible for every nonroot head and for the flat predictor. Intermediate supervision and total parameter counts still differ. These comparisons evaluate complete recipes rather than isolating architecture with identical supervision.

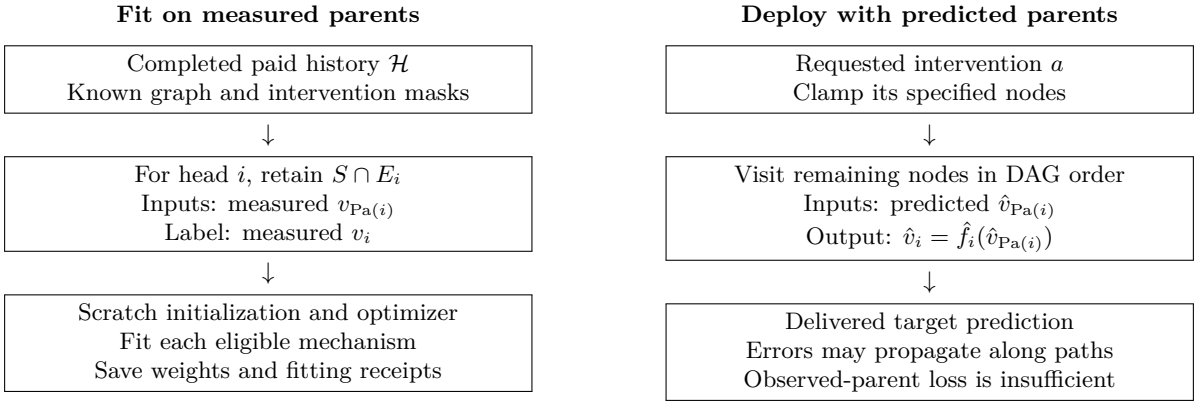


Figure 1: The structured delivery procedure separates mechanism fitting from free-running deployment. Rows intervened on head i are ineligible for that head; an empty eligible set cannot fit its mechanism. The prospective experiment clamps roots only and seals every fit before generating held-out responses. The flat control instead fits a direct root-to-target map on the same prospective history, without intermediate labels. Additional optimization and intermediate supervision therefore differ across recipes. This schematic supplies no extrapolation, identification or error-bound certification.

Figure 1 summarizes the fitting and deployment interfaces. Each head's training input is an observed parent vector; its deployed input can differ because upstream nodes are predicted. Saving the completed history makes a new fit possible, but does not resolve that support mismatch.

Endpoints. For a target Y and an evaluation action set $\mathcal{A}$, continuous MSE is $|\mathcal{A}|^{-1} \sum_{a \in \mathcal{A}} (\hat{y}(a) - y(a))^2$. Prospective normalized MSE (NMSE) divides this quantity by the target variance in the corresponding training history, using no held-out responses to set the scale. Archived attribution instead normalizes by the population target variance of its exposed grid, shared across arms; physical errors use each condition's training target variance. Exact-level error is the fraction with $Q(\hat{y}) \neq Q(y)$ for the fixed archived quantizer. It extends the recorded level lattice beyond both endpoints using the adjacent end spacing and resolves midpoint ties downward, rather than clipping predictions to endpoint levels. Nonfinite predictions are invalid for scientific acceptance. It is secondary for genuinely quantized targets outside the archival confirmation; it is not interchangeable with continuous prediction error. We distinguish exposed-grid assessment, which includes acquired inputs, from action-block holdout evaluation. Table 1 summarizes the endpoints, reporting units and initialization rules; these distinctions prevent pooling histories, systems and physical conditions into a single evidence population.

3 Related work

Causal models and invariance. Neural causal models connect expressive function classes to structural causal inference, but expressiveness does not itself establish learnability or identification (Xia et al., 2021). Our setting supplies the graph and measured parents rather than recovering causal structure from unrestricted observations. Invariant prediction uses stable relationships across experimental settings for causal inference (Peters et al., 2016). Here invariance is an assumption supporting eligible-row regression; we do not perform invariant-prediction structure selection or claim its confidence guarantees. Work on causal representation learning addresses learning causal variables and mechanisms from observations (Schölkopf et al., 2021). We assume the variables are already measured, so representation discovery is outside the study.

Online learning and final prediction. Obtaining a final predictor from sequential learning has a substantial literature. Online-to-batch results can extract predictors with risk bounds from online hypotheses under specified sampling assumptions (Cesa-Bianchi et al., 2004). Our procedure instead fits anew on a retained interventional history and measures the data-use and compute changes. Its adaptive histories and neural optimization do not inherit those bounds. Batch refitting and replay are existing tools; the con-

Study	Reporting unit	Evaluation and scale	Initialization summary
Confirmation	Paired histories of one fixed emulator	Previously exposed discrete grid; snapped exact-level error	Median of three scored initializations; sensitivity summary
Attribution (A)	The same archived histories; exploratory contrasts	The same exposed grid; continuous NMSE using grid target variance	Initialization zero primary; one and two for sensitivity
Prospective (B)	Independent system within each specified family; two histories paired within system	Reserved joint root-action blocks; continuous NMSE using the corresponding training-history target variance	Initialization zero for deployment; one and two for sensitivity
Physical (C)	Condition-specific assessment of one apparatus; no independent-world tests	Held-out joint angle blocks; continuous NMSE using each condition's training target variance	Initialization zero for neural fits; fixed linear controls

Table 1: The studies assess different endpoints, evaluation supports and reporting units. Scored initialization summaries are not deployable selection rules. Conditions, rows and repeated initializations do not supply additional independent systems. Prospective performance remains unreported pending supplemental reproduction qualification.

tribution here is the controlled account of which rows, fitting budgets and deployment semantics explain the observed delivery behavior. GDumb already stores streaming examples and trains from scratch at test time (Prabhu et al., 2020); experience replay retains earlier samples during continual learning (Rolnick et al., 2019). Thus separating collection from scratch fitting is established. Our specific assessment concerns intervention eligibility, the difference between measured-parent fitting and predicted-parent deployment, admitted-versus-paid response accounting, and constrained recipe comparisons, rather than a new temporal fitting principle.

Prediction-dependent inputs and physical validation. Sequential prediction can fail when an earlier prediction changes later inputs; imitation-learning reductions address this distribution mismatch (Ross et al., 2011). A DAG surrogate similarly receives predicted parents at deployment, but we do not apply that algorithm or transfer its regret results to our intervention setting. Our deterministic bound states the needed support and Lipschitz assumptions directly. Causal Chambers supply manipulable physical systems with published causal models (Gamella et al., 2025). We use one archived light-tunnel task as a prediction check against physical and Fourier bases, rather than claiming a new apparatus or a live action-design experiment.

4 Completed delivery confirmation

The comparison retains the originally registered 12 histories of a single deterministic emulator. The delivery procedure fits separate mechanism MLPs from all eligible paid rows using full-batch Adam and additional epochs. Original online weights remain unchanged. Evaluation runs the learned chain on the full previously exposed discrete grid using predicted intermediate values. Acquired input points occur within this grid, so it is not a held-out prospective sample.

The primary endpoint is exact-level error after mapping predictions to the nearest target level. For each history, the registered numerator is the median of three scored optimization initializations, and the denominator is the original online chain's error. That median summarizes sensitivity; it is not a deployable rule for selecting a model without access to true test outcomes. Histories, not initialization repeats or grid rows, are the paired units.

The geometric mean paired ratio is 0.183, with a log-scale t interval of [0.102, 0.327] and exhaustive two-sided sign-flip $p = 0.00146484$. 11 of 12 histories improve. The worsening history 124753321 changes from 0.196339 online error to 0.395658 delivery median error (ratio 2.015177); it is retained without exclusion or rescue tuning. The registered numerical floor is below every observed paired error and does not alter these

ratios. The interval is conditional on the history sampling for this fixed emulator; interpreting the sign-flip test requires symmetry of the paired log differences under its null. The generated values and source hashes are listed in `claim_index.json`.

This supports the bundled delivery procedure on repeated histories of that emulator. Paid histories are matched, but fitting compute is not. It establishes neither acquisition superiority, causal-factorization attribution, a foundation model benefit, nor generalization over independently parameterized systems.

5 Exploratory attribution of the delivery benefit

5.1 Design and controls

We compare three unique-row data sets: the final 50-entry buffer, all rows actually admitted at live online updates, and all eligible paid rows. Each is fit from scratch for 100 or 30,000 epochs with fixed architecture, initialization scheme, optimizer and intervention masks. The original online model is a separate comparator. Flat MLPs have both equal-epoch and separately labeled matched-CPU comparisons. Fixed ridge, polynomial and tree controls are configured on an earlier development history, with no configuration tuning on the twelve exposed histories. Input ranges are recomputed from each row set's eligible observations. Retention contrasts therefore change both rows and training-derived scaling; identical parameter seeds need not give identical initial functions in physical coordinates. The all-paid 100/30,000 contrast holds the rows and scaling fixed.

Replay reconstruction reveals additional differences from the online training path: the selected observation is duplicated in consolidation; per-node fast adaptation can include a directly intervened node; and the final observation refresh enters the saved buffer after the last fit. These differences are recorded rather than silently attributed to data quantity. The scratch factorial controls masks among its cells but does not exactly replay the online objective. No selected clamped-node fast updates occurred in these archived confirmation histories, so that code-path limitation is not an observed explanation of their delivery improvement. Diagnostics retain support coverage, observed-parent residuals, propagated prediction shifts and decision margins, especially for the worsening history. No simulator responses are acquired in Stage A.

5.2 Results and interpretation

The full registered matrix completed and passed an independent audit of source, inputs, eligibility masks, original artifacts, fitted-model hashes and all score files. With initialization zero, all-paid SCM fitting for 30,000 epochs gives geometric mean continuous NMSE ratios of 0.522 versus unchanged online weights, 0.551 versus the development-selected flat refit, and 0.423 versus the separately matched-CPU flat fit. These are descriptive results from the previously exposed grid, not prospective superiority tests.

Within the scratch SCM factorial, the all-paid long fit has ratios of 0.096 versus final-buffer long fitting and 0.034 versus all-paid 100-epoch fitting. Its ratio versus online-admitted long fitting is 0.980: there is little aggregate continuous-error change, but 10 histories improve and individual ratios range from 0.644 to 2.874. The snapped-error ratio is 0.469, showing that the endpoint also matters. These are conditional scratch-fit contrasts, not an additive decomposition of the gain over online weights, which retain information and optimizer states from earlier updates. The registered exploratory rule selects the all-paid SCM at initialization zero, the frozen flat comparator and the short-fit optimization ablation for prospective evaluation.

SCM and flat fits share the paid history but differ in which intermediate-clamp rows can enter their regressions. The structured recipe can reuse measurements that a roots-only observational predictor cannot represent. Accordingly this comparison does not isolate a pure architecture effect with identical usable rows, nor prove general causal identifiability. Matched fit CPU includes every SCM head, including the communication-cost branch that does not feed the scored target. The complete matrix consumed 10.87 fit CPU-core-hours; this excludes process startup, evaluation and imports completed before each head's timer. Classical-library imports within the timer remain included. Table 2 retains all primary configurations; initialization sensitivity is preserved separately. SCM/flat continuous ratios for initializations one and two are 0.458 and 0.519; neither changes the primary initialization-zero rule. The CPU-budgeted flat compari-

son tests that specified training schedule, not the best prediction attainable by an optimally allocated flat compute budget. Table 5 retains each history.

Data	Learner	Epochs	NMSE/online	Snap/online	CPU h
Final50	SCM	100	26.889	4.082	0.002
Final50	SCM	30,000	5.466	2.593	0.390
Final50	flat	100	29.230	4.126	< 0.001
Final50	flat	30,000	29.158	4.103	0.078
Admitted	SCM	100	16.661	3.865	0.003
Admitted	SCM	30,000	0.533	0.435	0.783
Admitted	flat	100	13.198	3.695	0.001
Admitted	flat	30,000	2.395	1.949	0.159
All paid	SCM	100	15.448	3.846	0.006
All paid	SCM	30,000	0.522	0.204	1.455
All paid	flat	100	12.248	3.655	0.002
All paid	flat	30,000	0.948	0.950	0.270
All paid	ridge	–	28.890	4.150	0.002
All paid	polynomial	–	4.440	2.737	0.002
All paid	tree	–	24.582	3.883	0.004
All paid	flat	CPU	1.234	1.206	1.455

Table 2: Exploratory primary initialization0 across the twelve archived histories. Ratios are geometric means relative to unchanged online weights; lower is better. CPU hours sum fit process time across the twelve fits in each row, excluding evaluation and imports performed before the fit timer. Classical-library imports inside the timer remain included. SCM cost includes all five heads. CPU denotes the matched-CPU flat fit, whose update count varies. Epoch counts apply only to neural fits. All configurations and sensitivity initializations are retained in the receipt.

5.3 Retained failure and propagation

History 124753321 still worsens under the primary delivery fit, with snapped error 0.397. Its target mechanism's observed-parent MSE is 8.07×10^{-5}, while free-running chain MSE is 2.86×10^{-4}; the propagated prediction-shift MSE is 2.09×10^{-4}. Only 0.020% of true evaluation parent vectors lie outside the training bounding box. That summary establishes neither joint support nor coverage of deployed predicted-parent vectors. Writing $r = \hat{f}(p_{\text{true}}) - y$ and $d = \hat{f}(p_{\text{pred}}) - \hat{f}(p_{\text{true}})$, the grid average satisfies $\text{MSE}_{\text{chain}} = \mathbb{E}[r^2] + \mathbb{E}[d^2] + 2\mathbb{E}[rd]$. The cross-term prevents additive attribution of local residual and propagated shift. Neither this diagnostic nor the snapped endpoint identifies a unique cause. The history is retained without tuning a rescue or selecting a different scored initialization.

6 Frozen prospective evaluation

Result integration pending. The experiment and its original independent audit are complete. This section describes the released protocol, frozen before collecting responses or opening held-out outcomes. Supplemental reproduction qualification and integrated reporting remain outstanding, so this version supplies no prospective performance claim. The recipe, simpler control and short-fit ablation were selected at the completed attribution gate and are unchanged.

6.1 Systems, histories and estimand

We use 20 independently parameterized systems in each of two graph-size strata. The five-node family is an explicit parameterized extension of the audited legacy equations, whose unparameterized base is fixed; varying collection seeds alone would not make independent systems. The thirty-node family reuses the audited graph and coefficient generator with both random states pinned. These are controlled extensions within the specified families, not an external benchmark or a representative sample of all SCMs. Appendix B

gives the mechanisms and action construction. The five-node target depends effectively on one root; its other branch is irrelevant to that target. The families differ in structure and functions as well as node count. Graph size labels strata and does not isolate a scaling effect.

Each system supplies two shared 400-response histories: balanced varied-value root interventions and matched random root interventions. Each also has one shared set of 400 held-out root actions. Complete joint-action blocks are reserved before collection and cannot appear in either training history. All arms receive the same paid history. This design tests dependence on the collection strategy without establishing acquisition superiority. The target is X_3 in the five-node family and X_{30} in the thirty-node family.

Collection and evaluation both disable noise and target the same deterministic structural map. This estimand does not substitute a composed conditional mean for the expected outcome of a noisy nonlinear SCM. The first 50 paid training responses provide a common, fixed input calibration for structured learners before their retrospective updates. Thus the rolling comparator reproduces the archived online update kernel after calibration; it is not a streaming algorithm that operates before those calibration observations are available. The primary NMSE denominator uses the full training-history target variance.

6.2 Comparators and separation of fitting from evaluation

The primary comparison uses initialization zero for delivery, the original rolling update kernel, the selected flat refit and the 100-update SCM ablation. Initialization one and two fits of delivery and flat measure optimization sensitivity separately; no scored initialization is selected for deployment. There are eight fits per history, totaling 640 fits across the frozen systems. All model states, optimizer accounting, inputs and attempt receipts must be sealed before the evaluator generates any held-out response. Failed or interrupted attempts remain in custody and block acceptance; they are not replaced by favorable worlds or removed from the analysis.

Joint root interventions leave every nonroot mechanism eligible. Nevertheless, SCM and flat learners differ in intermediate supervision and total fitting compute. The comparison therefore tests the selected recipe against a simpler predictor, not an architecture effect at equal supervision or compute. Root-only histories can leave parent support on a lower-dimensional manifold, so this study also does not establish identification for arbitrary internal interventions.

6.3 Registered primary analysis

For system s, history h, and control c, define

$$z_{s,c} = \frac{1}{2} \sum_{h \in \{\text{balanced,random}\}} \left[\log \max(\text{NMSE}_{s,h,\text{delivery}}, \eta) - \log \max(\text{NMSE}_{s,h,c}, \eta)\right], \qquad \eta = 10^{-12}. \tag{2}$$

The numerical floor and its activation count are reported. The independent unit is the system, with histories weighted equally within it. Each stratum's ratio is $\exp(\bar{z}_c)$ with a marginal 95% t interval on the mean log ratio and a two-sided paired t test across its 20 systems. Holm adjustment covers four primary tests: delivery versus rolling and delivery versus flat in each of the two strata. An asserted superiority requires all three frozen conditions: ratio at most 0.8, upper marginal interval below one, and Holm-adjusted $p < 0.05$. The intervals themselves are not simultaneous intervals. These parametric intervals and tests rely on system-level log-ratio sampling assumptions; they are not distribution-free guarantees. Ablation, initialization sensitivity, history-specific summaries and failure diagnostics are reported separately. No grid rows, histories or optimization repeats are counted as independent systems. No result is accepted from an incomplete primary matrix.

7 Archived physical prediction

We use the eleven previously unscored conditions of `lt_malus_v1`; `white_64` remains development only. Duplicate commands and joint action blocks are grouped before splitting. Compare delivery, rolling-buffer learning, joint Fourier regression and the published relative-angle physics model, using continuous prediction

error. The physics basis is an intercept and $\cos^2(\theta_1 - \theta_2)$, with degree inputs converted to radians; the earlier product-form attenuation control is not retained. Conditions and repeated readings from one apparatus are not independent physical worlds. All conditions and conditional uncertainty are reported in Tables 3 and 4. At fixed initialization zero, delivery has lower row-weighted NMSE than rolling-buffer fitting in 7 conditions, than the physics fit in 3, and than Fourier regression in 0. These are descriptive counts, not independent-world tests. The study consumed 0.053 fit/evaluation CPU-core-hours, excluding startup and imports completed before the phase timers, and acquired no physical responses. All fits were sealed before held-out losses were computed. Fourier regression outperforming delivery in every condition rules out an external advantage over that control here; the fixed domain-informed control provides better predictions. This case contains one measured response mechanism, so it is not a test of DAG factorization. We retain the result without changing the recipe after scoring.

Condition	Delivery	Rolling	Physics	Fourier
blue_128	0.359	0.156	0.058	0.049
blue_255	0.012	0.017	0.026	9.43×10^{-4}
blue_64	0.471	2.087	0.103	0.102
green_128	0.789	0.290	0.058	0.046
green_255	0.010	0.038	0.018	0.002
green_64	5.167	3.307	0.121	0.102
red_128	0.276	0.389	0.047	0.045
red_255	0.011	0.063	0.024	0.004
red_64	1.119	0.823	0.159	0.135
white_128	0.163	0.462	0.044	0.042
white_255	0.020	0.023	0.006	0.002

Table 3: Continuous held-out MSE divided by the training variance for all eleven evaluation conditions, excluding development condition `white_64`. Lower is better. These conditions are readings of one apparatus, not independent systems.

Condition	Delivery/Rolling	Delivery/Physics	Delivery/Fourier
blue_128	2.55 [0.50, 8.03]	6.61 [0.97, 17.08]	7.88 [1.40, 17.91]
blue_255	0.71 [0.24, 1.01]	0.47 [0.10, 1.22]	13.05 [2.29, 28.48]
blue_64	0.22 [0.07, 1.12]	4.38 [1.56, 8.49]	4.41 [1.56, 8.29]
green_128	2.58 [0.43, 4.38]	12.96 [1.15, 38.54]	16.22 [1.91, 46.75]
green_255	0.23 [0.06, 2.40]	0.54 [0.10, 1.57]	6.26 [1.41, 19.05]
green_64	1.56 [0.17, 1.78]	43.05 [1.14, 125.29]	51.08 [1.51, 136.39]
red_128	0.75 [0.08, 2.97]	6.47 [1.04, 14.30]	6.64 [1.21, 13.20]
red_255	0.18 [0.06, 0.60]	0.52 [0.18, 1.36]	2.74 [1.48, 4.21]
red_64	1.31 [0.19, 4.59]	6.54 [1.40, 16.18]	7.37 [1.73, 18.01]
white_128	0.36 [0.12, 1.12]	3.86 [1.09, 7.31]	3.95 [1.16, 7.86]
white_255	0.85 [0.16, 3.55]	2.92 [0.54, 6.83]	8.57 [1.65, 15.65]

Table 4: Equal-action-block error ratios and conditional 95% bootstrap intervals. Eight held-out blocks are resampled within each condition. These ratios weight blocks equally, whereas Table 3 weights held-out rows equally. Intervals describe uncertainty conditional on this apparatus and observed blocks; they are not independent-world confidence intervals or multiplicity-adjusted superiority tests.

8 Conditions and limits of delivery

We study a known acyclic graph with measured parents. The following statements explain permissible reuse and possible error amplification. They do not prove identification for arbitrary neural models or certify the fitted networks.

Proposition 8.1 (Mechanism eligibility)**.** *Let $V_i = f_i(V_{\mathrm{Pa}(i)}) + U_i$ be an invariant additive-noise mechanism. In each experiment, suppose the current exogenous noises are independent across mechanisms and independent of all past experiments; U_i is integrable and has mean zero. An intervention policy may depend on previous experiments, but must choose its action before the current exogenous noises are realized. Retain a row only when the action does not replace mechanism i, and let S_i denote the final retained-row indicator, including any terminal selection. Require $\mathbb{E}[U_i \mid V_{\mathrm{Pa}(i)}, A, H, S_i = 1] = 0$ on the retained support, where A is the current action and H the preceding history. Conditional independence of S_i and U_i given parents, action and history is a sufficient retention condition. Then, in the retained mixture,*

$$\mathbb{E}[V_i \mid V_{\mathrm{Pa}(i)} = p, \textit{retained}] = f_i(p).$$

Thus squared-loss regression can target this mechanism on supported parent values, assuming finite second moments. This is an estimand statement; finite data, optimization and extrapolation guarantees require further assumptions.

Proof. In an acyclic graph, a node's parents depend only on exogenous noises of preceding nodes in topological order and the chosen action. The current U_i is independent of those noises and of past experiments. The policy precedes the current realization, and the stated retention condition preserves conditional mean zero. Substituting the structural equation therefore gives $\mathbb{E}[V_i \mid V_{\mathrm{Pa}(i)}, A, H, S_i = 1] = f_i(V_{\mathrm{Pa}(i)})$. Taking the further conditional expectation over action and history yields the assertion. The conditional expectation minimizes population squared loss on the retained support. No statement outside that support follows. □

Confounding, temporal dependence of current noise on prior experiments, missing parents, interventions that change rather than clamp a mechanism, and response-dependent retention can invalidate the premise. In particular, selecting the best probe using its measured response can change the noise distribution. All-paid reuse avoids discarding probes on that basis, but still requires invariant mechanisms and measured parents. The current confirmation uses a deterministic emulator; this distinction matters for any extension to noisy systems. Even within this additive-noise class, composing conditional means need not produce a downstream interventional mean through a nonlinear mechanism. For example, clamp $W = 0$, set $X = W + U$ with $\Pr(U = 1) = \Pr(U = -1) = 1/2$, and let $Y = X^2$. The conditional-mean mechanism for X gives zero at $W = 0$, so composing the mean mechanisms predicts zero for Y. The actual interventional expectation is one. The zero-noise structural map and the noisy interventional mean are distinct estimands, even when both conditional-mean mechanisms are exactly known.

Proposition 8.2 (Deterministic DAG propagation)**.** *Consider a deterministic SCM and its learned functions $\hat{f}_i$, evaluated in topological order under identical correctly clamped interventions. For each non-clamped node, assume $|\hat{f}_i(v_{\mathrm{Pa}(i)}) - f_i(v_{\mathrm{Pa}(i)})| \le \epsilon_i$ at every true parent vector in the deployment support. Assume also*

$$|\hat{f}_i(p) - \hat{f}_i(q)| \le \sum_{j \in \mathrm{Pa}(i)} L_{ij} |p_j - q_j|$$

on a region containing both true and predicted parent vectors, with $L_{ij} \ge 0$. Writing $e_i = |\hat{v}_i - v_i|$, one has

$$e_i \le \epsilon_i + \sum_{j \in \mathrm{Pa}(i)} L_{ij} e_j.$$

Correctly clamped nodes have $e_i = 0$. If their rows in L and their local bounds are set to zero, the vector bound is $e \le (I - L)^{-1}\epsilon = \sum_{k=0}^{d-1} L^k \epsilon$, where d is the number of nodes and L is strictly triangular in topological order.

Proof. Add and subtract $\hat{f}_i(v_{\mathrm{Pa}(i)})$ and apply the triangle inequality. The local approximation bounds one term; the coordinate Lipschitz inequality bounds the other. Clamps are identical by assumption. Substitution in topological order gives the path sum. A strictly triangular $d \times d$ matrix satisfies $L^d = 0$, so the finite geometric sum is its inverse. □

The bound can amplify a small local error along deep or high-gain paths. Observed-parent test MSE is neither a uniform ϵ_i nor a verified Lipschitz constant. Our residual and support diagnostics are explanatory measurements, not certified instances of this proposition.

Proposition 8.3 (Root interventions need not constrain propagation)**.** *Consider the known graph $R \to M$, $R \to Y$, $M \to Y$, with $M = R$ and $Y = f(R, M) = R + M$. For any $K \notin \{-1, 1\}$, the alternative mechanism $g_K(R, M) = R + M + K(M - R)$ uses both declared parents and agrees with f under every intervention on R alone. If a learned upstream mechanism instead predicts $\hat{M} = R + \delta$, its free-running prediction using g_K has error $|(1 + K)\delta|$, although its observed-parent fit is exact.*

Proof. Root interventions leave the parent support of Y on the diagonal $M = R$, where the added term vanishes. Substitution of $\hat{M}$ gives $g_K(R, \hat{M}) = 2R + (1 + K)\delta$, whereas the true outcome is $2R$. Both parent coefficients are nonzero for the stated choices of K. □

This failure need not leave a coordinate bounding box. For example, take $R = 0.25$, $\delta = 0.01$ and $K = 1000$. Both true and predicted parent vectors are inside $[-1, 1]^2$, but the prediction error is 10.01. Observations covering the diagonal cannot constrain sensitivity normal to it; more root observations alone do not remove that ambiguity. An internal intervention can distinguish these mechanisms. Our prospective root-only protocol assesses prediction under held-out root interventions, not unrestricted internal-action identification. This is an existence example, not a diagnosis of a measured failure or a verified bound for the fitted networks.

Proposition 8.4 (Quantization margin)**.** *For $m \geq 2$ distinct ordered levels $\ell_1 < \cdots < \ell_m$, let Q assign the nearest level, with a fixed tie rule, and let $b_k = (\ell_k + \ell_{k+1})/2$. Define $M(y) = \min_k |y - b_k|$. If $|\hat{y} - y| < M(y)$ then $Q(\hat{y}) = Q(y)$. If a pointwise prediction bound $B(y)$ is valid, $\Pr(Q(\hat{y}) \neq Q(y)) \leq \Pr(M(y) \leq B(y))$ under any specified deployment probability measure for which the bound holds almost surely.*

Proof. The open interval of radius $M(y)$ around y crosses no decision boundary. Consequently both values occupy the same decision cell. The contrapositive implies that a quantization error requires $M(y) \leq |\hat{y} - y| \leq B(y)$. Strict inequality excludes boundary ties. □

Snapped error depends on the location of errors relative to decision boundaries; similar MSE alone need not imply similar exact-level accuracy. This finite-level example clips outside values to endpoint cells. The empirical scorer instead extends the lattice past both endpoints. The same argument uses the complete set of boundaries of that extended quantizer; at a recorded true level, its nearest boundary distance remains half the adjacent gap. The empirical evaluation measure is uniform over the archived grid.

Proposition 8.5 (Squared error and snapping risk)**.** *Under a specified deployment probability measure, let $R = \mathbb{E}[(\hat{Y} - Y)^2] < \infty$ and use the same quantizer and margin M as above. For every $\tau > 0$,*

$$\Pr(Q(\hat{Y}) \neq Q(Y)) \leq \min\{1, \Pr(M(Y) \leq \tau) + R/\tau^2\}.$$

If $M(Y) \geq m > 0$ almost surely, the sharper bound is $\Pr(Q(\hat{Y}) \neq Q(Y)) \leq \min\{1, R/m^2\}$.

Proof. A snapping error requires $|\hat{Y} - Y| \geq M(Y)$. Its event is therefore contained in the union of $\{M(Y) \leq \tau\}$ and $\{|\hat{Y} - Y| > \tau\}$. Apply the union bound and Markov's inequality to the squared error. Under the positive-margin assumption the error event instead requires $(\hat{Y} - Y)^2 \geq m^2$, giving the second bound directly. □

For a uniformly spaced lattice with gap Δ and true responses on its levels, $m = \Delta/2$. The second inequality also applies to a finite evaluation grid by taking its empirical measure and continuous MSE; it may be vacuous. It is a consequence of evaluated error and quantizer geometry, not a guarantee on new actions.

For continuously valued targets, concentration near decision boundaries can make even small squared error compatible with frequent snapping errors. Neither bound orders two models by snapped error from their MSE alone.

Counterexamples. For a constant predictor and test response zero, fitting a single training response zero gives zero risk; adding a noisy response two makes its least-squares mean one and test risk one. With a noisy training target one, optimizing a constant from 0.1 to one reduces training error but increases test error. For a two-stage chain, suppose the true parent and child are zero on the observed support, the learned parent predicts 0.1, and the learned child is $\hat{f}(z) = 100z$. The child has zero observed-parent error at zero, but its free-running prediction is ten. These are existence examples, not descriptions of the measured histories. Analytically solved chains, forks, colliders and clamps are checked by executable tests.

9 Reproducibility and compute accounting

Fitting and evaluation run in separate processes. Training inputs, row selections, source, dependency versions, optimizer configuration and outputs are hash-bound. Stage A fit workers are denied network access and evaluation arrays. The physical worker uses only registered training rows for fitting and normalization and seals predictions before computing held-out losses. Exclusive output paths preserve interrupted attempts and reject overwriting accepted artifacts. Historical confirmation continuation is disclosed below; prospective system replacement is forbidden. Supervisors enforce wall-time and memory ceilings. Eligible rows, parameters, optimizer updates and CPU time accompany fits. Archived attribution records fit timing/RSS; prospective fits add supervised attempt memory. Physical memory and elapsed telemetry are available at whole-process or phase granularity, rather than independently for each method. Original histories and weights are retained under local result custody. The claim generator binds every retained numerical result to its validated receipt. Experimental controls, analytic checks and the execution ledger are versioned with the manuscript. No closed-source API is needed for the critical path.

10 Discussion and limitations

Data retention and fitting effort. The exploratory contrasts distinguish keeping a completed history from keeping only the final buffer, and a short scratch fit from a much longer fit. Both changes are consequential within the scratch-fit matrix. Retention also changes normalization, while the online comparator carries persistent weights and Adam states; this matrix does not isolate their separate causal effects. The all-paid versus admitted-row aggregate continuous contrast is small, but has marked history-level and endpoint heterogeneity. This distinction is useful for experimental systems: recording observations and budgeting a final fit can improve delivery without a change in acquisition. It does not imply that more data or more updates improve every system, as the retained failure and counterexamples show.

Structured prediction and support. A known graph gives a decomposition and a rule for row eligibility. It also creates a deployment dependency: upstream error changes the inputs of later heads. A low observed-parent residual cannot by itself validate the deployed chain, and a coordinate bounding box cannot establish joint support. Intermediate-intervention measurements may constrain mechanisms that a root-only predictor cannot learn from those inputs, but no pure factorization advantage follows from a comparison with unequal usable supervision. The prospective protocol deliberately tests prediction of root-interventional maps; its results will not resolve the internal-action ambiguity.

External boundary and reporting units. The physical task is a single mechanism in one apparatus. Neural delivery wins some comparisons with a rolling learner, but loses every comparison with the joint Fourier predictor. This limits claims about a generally preferable neural recipe and gives a concrete example where domain structure is more useful. Condition-specific bootstrap intervals describe observed action blocks, not a population of independently built apparatuses. Similarly, the archived histories supply conditional evidence about one emulator, not twelve causal worlds. The prospective systems address variation within two generator families, with broader physical generalization still outside their scope.

Remaining evidence and validity limits. The archived evaluation grid was previously exposed and includes acquired inputs. Attribution on that grid is exploratory, even though its configurations were fixed before the new fits. The confirmation's median of three scored initializations describes sensitivity and must not be read as a deployable model-selection rule. The prospective fixed-initialization protocol is needed to test the selected recipe on unused action blocks. Compute totals distinguish fitting process time from full execution and requested allocations; an equal-epoch neural comparison is not an equal-compute comparison. The eligibility and propagation propositions require measured parents, invariance, noise and support assumptions that fitted residuals cannot certify. Causal discovery, counterfactual identification and foundation-model capability are not tested.

11 Conclusion

A completed experimental history can support a substantially different delivered predictor from the learner maintained during collection. Within the archived scratch-fit matrix, long fits on retained histories improve substantially over final-buffer and short fits, while unused paid rows have a smaller aggregate continuous effect with substantial heterogeneity. This is a conditional recipe comparison, rather than a partition of the online-to-delivery gain. Structured delivery remains favorable against the selected simpler refits on the exposed grid, while a physical prediction study shows a clear boundary against domain-informed controls. The theoretical analysis explains why eligible-row fitting, joint support and free-running error must be considered separately. The frozen prospective comparison has completed its original independent audit; supplemental reproduction qualification and integration remain outstanding. Until those reporting gates pass, the empirical conclusion in this version is restricted to the completed archived studies.

References

Nicolò Cesa-Bianchi, Alex Conconi, and Claudio Gentile. On the generalization ability of on-line learning algorithms. *IEEE Transactions on Information Theory*, 50(9):2050–2057, 2004. doi: 10.1109/TIT.2004.833339. URL https://cesa-bianchi.di.unimi.it/Pubblicazioni/J20.pdf.

Juan L. Gamella, Jonas Peters, and Peter Bühlmann. Causal chambers as a real-world physical testbed for ai methodology. *Nature Machine Intelligence*, 7:107–118, 2025. doi: 10.1038/s42256-024-00964-x.

Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: Identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78(5):947–1012, 2016. doi: 10.1111/rssb.12167.

Ameya Prabhu, Philip H. S. Torr, and Puneet K. Dokania. GDumb: A simple approach that questions our progress in continual learning. In *European Conference on Computer Vision*, 2020. URL https://www.ecva.net/papers/eccv_2020/papers_ECCV/papers/123470511.pdf.

David Rolnick, Arun Ahuja, Jonathan Schwarz, Timothy P. Lillicrap, and Greg Wayne. Experience replay for continual learning. In *Advances in Neural Information Processing Systems*, 2019. URL https://arxiv.org/abs/1811.11682.

Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pp. 627–635, 2011. URL https://proceedings.mlr.press/v15/ross11a.html.

Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio. Toward causal representation learning. *Proceedings of the IEEE*, 109(5):612–634, 2021. doi: 10.1109/JPROC.2021.3058954. URL https://arxiv.org/abs/2102.11107.

Kevin Xia, Kai-Zhan Lee, Yoshua Bengio, and Elias Bareinboim. The causal-neural connection: Expressiveness, learnability, and inference. In *Advances in Neural Information Processing Systems*, 2021. URL https://arxiv.org/abs/2107.00793.

A Implementation and accounting details

The neural architecture for an input dimension d is Linear$(d, 64)$–ReLU–Linear$(64, 64)$–ReLU– Linear$(64, 1)$. Each head therefore has $64d+4289$ trainable parameters; structured-model totals sum the head counts rather than counting only the scored target head. The retrospective scratch fits use separate Adam optimizers with constant learning rate 0.002, full batches and squared loss. Neural fit receipts include optimizer update count, row IDs, parameter count and CPU process time; memory is measured at fit/attempt or whole-process granularity as specified above. An epoch denotes one full-batch update per fitted head, not an acquired observation.

A.1 Archived emulator and acquisition histories

The archived task is a hand-coded deterministic emulator, rather than a physical world. It has eight controllable roots and five measured nonroot mechanisms: two link scores L, M, their combined score I, the target Y and a separate cost branch C. Write the roots as $(N, D, h, p, N_M, h_M, p_M, J)$, representing total and dynamic counts, altitude and power for the first link, count, altitude and power for the second, and shared jamming. Their five-level supports are

$$\begin{aligned} N &\in \{250, 350, 450, 550, 650\}, & D &\in \{0, 50, 100, 150, 200\}, \\ h &\in \{250, 300, 350, 400, 450\}, & p &\in \{1, 1.25, 1.5, 1.75, 2\}, \\ N_M &\in \{5, 10, 15, 20, 25\}, & h_M &\in \{6000, 7000, 8000, 9000, 10000\}, \\ p_M &\in \{1, 1.25, 1.5, 1.75, 2\}, & J &\in \{0, 0.25, 0.5, 0.75, 1\}. \end{aligned}$$

Let $s(z) = \min(1, \max(0, z))$ and $d = D/N$. The fixed mechanisms are

$$\begin{aligned} L &= s\Bigg(0.35s\left(\frac{N + 0.3D - 100}{600}\right) + 0.25s\left(1 - \frac{|h - 350|}{300}\right) \\ &\qquad + 0.40s\left(\frac{p(1 + 0.15d) - 0.8}{1.2}\right) - 0.30s(J(1 - 0.6d))\Bigg), \\ M &= s\left(0.25s\left(\frac{N_M - 5}{20}\right) + 0.30s\left(1 - \frac{|h_M - 8000|}{4000}\right) + 0.45s\left(\frac{p_M - 0.8}{1.2}\right) - 0.25s(0.85J)\right), \\ I &= s(0.55L + 0.45M), \qquad Y = \frac{\mathrm{round}_{\mathrm{even}}(30s((I - 0.35)/0.50))}{30}, \qquad C = N + 0.75D. \end{aligned}$$

The shared J is a fork and I is a collider; C never feeds the target. Response generation follows frozen Python rounding; scorer midpoint ties are separately defined above. The 31 target levels are $\{k/30 : k = 0, \ldots, 30\}$. Every one of the $5^8 = 390{,}625$ root actions receives equal weight in the exposed grid. Its target variance normalizes archived continuous errors.

Each independently seeded history runs 200 iterations with four random proposed probes per iteration and outcome-dependent mechanism selection, no model API or policy update. Frozen settings include three initial rows, 500 validation rows, 100 learner epochs, a 50-entry buffer and a 25-row observation refresh every five iterations. All paid calls, including unselected probes, are logged. Case seeds derive from the first 32 SHA-256 bits of a fixed pre-outcome case-index string modulo $2^{31} - 1$; the full seed rule and settings accompany the frozen protocol. Independence concerns these acquisition streams, not different mechanisms.

Archived attribution. The row sets use unique paid query IDs, not unique action values. The admitted set is the union available during actual live updates, excluding probe-clone training. Reconstruction accounts for the selected row's duplicated appearance in consolidation and for observation refreshes added after fitting. Intervention masking excludes a directly clamped response from its own mechanism fit. Inputs are normalized as $(x - \mathrm{lo})/(\mathrm{hi} - \mathrm{lo})$ using eligible training minima and maxima, without input clipping; a constant range is given width one. The structured fit includes the communication-cost branch even though it does not feed the evaluated target. In every archived history, the final buffer supplies 50 unique paid rows per head and

History	Online NMSE	SCM NMSE	Flat NMSE	SCM/admitted	SCM snap
27424209	0.002492	0.001273	0.002948	0.959	0.023
1726880744	0.003233	0.001782	0.002801	0.946	0.148
735595885	0.002856	0.001306	0.002708	0.877	0.017
983656467	0.003046	0.001319	0.002665	0.935	0.029
124753321	0.002734	0.004567	0.002534	2.874	0.397
921441405	0.002524	0.001342	0.002783	0.929	0.042
934168586	0.003245	0.001304	0.002946	0.854	0.020
546725957	0.002488	0.001339	0.002727	0.792	0.042
520888668	0.002959	0.001585	0.002682	1.091	0.116
1091699608	0.003617	0.001307	0.002751	0.644	0.019
585254818	0.002309	0.001301	0.002742	0.884	0.018
884825602	0.003601	0.001272	0.002688	0.939	0.020

Table 5: Every archived history at fixed initialization zero. SCM and flat draw their respective eligible subsets from the same paid history and use 30,000 full-batch updates per SCM head and for the flat model. A mechanism head excludes rows that clamp that mechanism; the root-to-target flat fit excludes rows that clamp any nonroot node. Admitted refers to the long SCM fit on the admitted-row union. NMSE uses the shared exposed-grid target variance. The final column is SCM exact-level error. These exploratory outcomes retain history124753321 and are not prospective tests.

the admitted union 1,178. All-paid mechanism fits have 4,303 eligible rows for the two links and combined score, and 4,803 for the target and cost; the root-only flat fit has 3,803. Counts refer to query identities, not distinct commands. The five-head structured model has 22,341 parameters; the eight-input flat model has 4,801. These unequal counts and supervision remain part of the recipe comparison.

Ridge uses penalty 1; polynomial regression uses degree-three features and ridge penalty 1; the extra-trees control uses 256 trees, minimum leaf size two, random state zero and one worker. The flat MLP uses the same width and optimizer as a mechanism head. Configurations and the simpler-control selection were fixed using the earlier development history. The matched-CPU flat control receives the sum of the corresponding all-paid long SCM heads' fit process time, excluding evaluation and imports completed before each head's timer. Its update count varies and is retained; it is not presented as another 30,000-update cell. The full matrix contains 480 fits, including the two budgets, three row sets and initialization sensitivity cells, plus the fixed classical and matched-CPU controls.

Original rolling comparator. Original archived weights are never refit in the confirmation or attribution comparison. In the prospective study, the archived training kernel is invoked directly with its persistent Adam states, 50-entry replay and duplicated newly selected observation. Its fast adaptation and consolidation updates remain unchanged. The known directly-clamped-node fast-adaptation limitation is inactive in the root-only prospective histories. The common calibration prefix avoids changing feature scales using held-out observations. After calibrating on the first 50 rows, retrospective online replay starts with an empty buffer and processes all 400 arrivals, including that prefix. Each arrival performs 20 fast and 100 consolidation updates per nonroot head, totaling 48,000 per head. Consolidation uses up to 50 buffer entries plus the duplicated current row. Neural computation uses float32 and the recorded Adam defaults; stage-specific dependency and seed schemes accompany the release.

B Prospective systems and action construction

The five-node family has roots X_1, X_4 and mechanisms

$$X_2 = aX_1 + b, \qquad X_3 = cX_1 - dX_2 + s\sin(X_2), \qquad X_5 = qX_4^2.$$

Independent coefficient draws multiply base coefficients $(a, b, c, d, s, q) = (2, 1, 0.5, 1, 1, 0.2)$ by independent uniform factors in $[0.8, 1.2]$. The thirty-node family has five layers of sizes $(5, 5, 10, 5, 5)$ and edge coefficients drawn uniformly in $[0.3, 0.7]$. Nodes in the second layer choose one or two roots uniformly without

replacement. In the third layer, nodes whose one-based index is divisible by three choose two second-layer parents; other nodes choose one. Fourth-layer nodes choose two or three third-layer parents uniformly without replacement. Each final-layer node chooses one fourth-layer parent. Graph and coefficient random states are pinned by the system descriptor; no edge or coefficient is redrawn per response. For a nonroot i, write $u_i = \sum_{j \in \mathrm{Pa}(i)} w_{ij} X_j$; its structural response is $u_i + 0.2 \sin(u_i)$ when its one-based node index is divisible by five, and u_i otherwise. Both families use zero exogenous noise in this study. All graph and coefficient descriptors precede response generation and are shared unchanged across arms.

Each root action lies in $[-3, 3]$. Joint blocks are Cartesian products of five width-1.2 bins, with the upper endpoint assigned to the last bin. A fixed hash of each block reserves approximately one fifth of blocks for evaluation. Random training and evaluation actions are uniform draws accepted only in their respective block partitions. Balanced training favors the least-used of eleven evenly spaced levels at each root, subject to the same block exclusion; the bounded tie and quota rules are frozen in the implementation. Duplicate commands remain charged observations. No observed outcome determines which action or world enters the frozen design.

Mechanism input ranges are fixed from the first 50 paid training rows; root ranges use the declared $[-3, 3]$ support. For initialization r, each nonroot head is seeded by r plus its position in the topological order; the flat head uses r plus the number of nodes. These deterministic offsets are shared across the corresponding structured arms. No outcome selects an offset, initialization or replacement world.

C Physical split and uncertainty

The archive task uses the two polarizer commands and the measured light response `vis_3`. Angles are in $[-90, 90)$ degrees. For command pair (θ_1, θ_2), define $b_j = \lfloor (\theta_j + 90)/30 \rfloor$. A joint block is held out if $(b_1 + 2b_2) \bmod 5 = 0$; identical commands and their repeated readings always share a block. The development condition `white_64` is excluded from the eleven-condition evaluation. Neural output centering and scaling use the training mean and standard deviation, and are inverted before scoring; neural angle inputs are divided by 90. The neural delivery fit uses 30,000 full-batch updates. Rolling fitting follows archive order, using the last 50 eligible training rows and 100 updates per arriving row with persistent Adam. Its retrospective access to the training normalizer is shared with delivery and is not a strictly streaming normalization claim.

Physics regression fits an intercept and $\cos^2(\theta_1 - \theta_2)$ after conversion to radians. Fourier regression uses the 25 tensor-product features from $\{1, \sin(2\theta), \cos(2\theta), \sin(4\theta), \cos(4\theta)\}$ at each angle. Both fit linear coefficients on training rows only. Conditional intervals use 2,000 paired bootstrap resamples of the eight held-out action blocks within each condition, with fixed bootstrap seed 620061. Resampling gives each block equal weight; the primary descriptive NMSE instead weights readings equally. This difference explains why a block-weighted ratio need not equal a ratio computed from the row-weighted table. Neither bootstrap establishes uncertainty across independently constructed physical systems. The intervals are a conditional resampling assessment under block exchangeability; archive-order dependence or unobserved action blocks can limit that interpretation.

D Historical execution and release accounting

The confirmation's accepted terminal audit retains 60,962 aggregate charged emulator attempts. The twelve complete histories contain 57,636 persisted responses; 3,326 additional journal entries belong to an interrupted acquisition attempt without a persisted response history. Journals reserve a charge before invocation, so the returned-response count for that interrupted attempt is unknown. Retained copies are counted once, and delivery refits acquire no additional responses. An explicit execution amendment retained the first eleven completed histories and the interrupted final attempt, then ran a separately retained final attempt for that same registered history before scoring the complete matrix. This continuation was not chosen by scored performance. Stage A and the physical study acquire no new responses: they reuse frozen observations. The confirmation worker measures refit durations using a monotonic elapsed-time clock. These intervals overlap the containing acquisition stages; the nested charged/reserved wall budgets are cumulative accounting records, rather than additional executions. Historical process CPU for the complete confirmation campaign

is unmeasured. We therefore report that quantity as unknown and do not infer a whole-project CPU total by adding elapsed time, reservations, fit process CPU and scheduler allocation. Replay, packaging and review measurements cover their identified runs only. Stage A records 10.87 fit CPU-core-hours and the physical study 0.053 fit/evaluation CPU-core-hours. These measurements exclude startup and imports completed before the respective timers; classical-library imports inside the attribution fit timers remain included. The prospective qualification failure charged 126 allocated CPU-seconds; the successful replacement charged 119. Both remain in the final allocation accounting. Requested reservation, process CPU and supervised elapsed time are different quantities. Final prospective costs remain pending and will be reported with the completed matrix and audit, rather than inferred from the reserved allocation or a partial checkpoint.

E Verification and evidence provenance

The retained claim index maps each empirical macro to a validated score receipt, source hashes and acceptance record. The archived confirmation is linked to its independently recomputed paired statistics; attribution is linked to its immutable selection gate, full matrix and eligibility audit; physical results are linked to the fit seal and eleven-condition acceptance. The prospective registration freezes world descriptors, actions, dependencies, initialization, normalization, primary contrasts and resource ceilings before responses. Release requires independent recomputation of the completed result custody and all declared primary contrasts. No prospective score is included in this version.

Executable analytical checks cover chains, forks, colliders, clamped nodes, root-support ambiguity, quantization margins and the nonlinear conditional-mean counterexample. Scientific checks also verify replay reconstruction, intervention masks, predicted-parent inference, action-block disjointness, initialization membership and system-level statistical calculations. Passing implementation checks verifies those constructed cases, not the assumptions of a general statistical guarantee. A reproducibility release must include frozen protocols, exact observation ledgers, model and optimizer receipts, interrupted-attempt records, compact outcomes and verification commands. Author paths and identifying metadata must be removed from the anonymous review copy; original custody artifacts remain unchanged. The derived anonymous release must use relative artifact locations and its own hashed manifest mapping to original immutable digests, rather than editing hash-bound source receipts. Its verification entry point must resolve local artifacts independently of author paths, include the archived learner source and reconcile split fit snapshots without replacing conflicts. An offline unrelated-directory verification remains a submission gate. Archived attribution audits bind checkpoints to scores but do not independently replay every checkpoint; the physical audit independently recomputes sealed prediction errors and physics predictions, but does not replay its neural or Fourier models. A subsequent, separately hashed release adapter reconstructs the retained physical neural checkpoint and linear coefficient predictions, continuous errors and conditional bootstrap statistics from relocated artifacts. A distinct attribution adapter reconstructs every retained factorial and control checkpoint's scores, mechanism diagnostics and quantization margins, including the worsening history, from the archived learner, paid rows and exposed grid. A confirmation adapter additionally reconstructs the original online models and three delivery initializations, the paired log-error statistics, and disjoint charged-attempt versus persisted-response counts, retaining the interrupted journal. These adapters run without optimization or response acquisition. Their receipts and code digests are recorded separately from the original frozen workers and acceptance records. A separate metadata checker reconstructs every current empirical macro and compares the generated LaTeX values with the bound claim index; this does not verify every prose statement or table. A distinct read-only table checker compares the attribution, per-history and physical tables' headers, row membership and displayed values with independently pinned accepted receipts, preserving the distinction between row-weighted errors and block-weighted ratios. This check excludes captions, prose and prospective results. Supplemental reconstruction does not establish model refitting, historical freeze timing or final reviewer anonymity. Prospective acceptance requires all 640 checkpoint replays. Python audit hooks restrict trusted worker access; they are not an operating-system sandbox. The physical fit process loads the archive and uses only its registered training indices for objectives and scales before sealing predictions and scoring.